

Analysis and Design of a Clothing Web-Based System(CWBS)

Term Project

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Tobi Bakare -14247736

Aisha Ali-Ahmad -14240311

Junaid Asim - 14458071

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CHAPTER_1: SYSTEM ANALYSIS

(1) The Problem Statement:

An in-person store is trying to create an online platform to reach out to more customers throughout the country and make more sales.

(a) Context and importance of the system:

In the age of the internet, it is important for an in-person store to go online to expand its market. To reach a wider range of customers, the store has decided to expand to an online market which allows them to sell more clothes to people all over the country. This will increase profit for them to also branch their in-person stores to different states. An online store also collects real time data such as items most looked at, items left in the cart. This will help the store to cater to their customers' needs.

(b) Overall goals of the system:

The overall goals of the system is to keep track of inventory, increase customers, and receive payments online.

(c) Scope of the project:

I. IN-Scope: CWBS will include customers, inventory and payments

II. OUT-Scope: CWBS will not include store activity such as purchases made at the store, store inventory and in-person payments.

(2) Requirements:

2.1 Functional Requirements:

The website will have the option of “checkout as Guest” or “log in” CWBS will perform the following functions:

1. View and search through different clothing
2. Sort through different categories
3. Put items in the cart and take items out of the cart
4. Take payments with only credit cards
5. Add/change and save shipping information
6. Join the rewards program for loyal customers

2.2 Data Requirements:

For customers, there will be an option to login, so that their information is saved next time they visit the website.

For each billable item, CWBS will keep track of the type of clothing, item number and price.

For each invoice from a customer, CWBS will keep track of invoice number, invoice date, total amount that is yet to be paid

For each payment, CWBS will keep track of payment date, item number, amount of item, shipping address and payment type

2.3 Business Rules and Data Logic:

1. Customers will only be able to add in-stock items to the cart.
2. The website places an order on the completion of the payment process.
3. Acceptable payment methods are American Express, MasterCard, Visa Discover, and PayPal Only.
4. The website stores the payment information of a customer on a storage system. A system that is encrypted and meets PCI data security standards.
5. The shipping is only offered domestically so far. Its cost is calculated on the checkout page based on the region or address. The order with a total of over \$100 before taxes is shipped free.
6. Taxes are calculated on the cart page based on the tax rate according to the customer's state.
7. Cancellation of a placed order is not possible once shipped.
8. Returns are offered with the option of refund or exchange within 30 days of purchase once items are delivered.
9. Other than customers, the business itself will also use the website to perform important activities like adding new items, updating the quantity of new inventory, and keeping track of new customers, etc.

2.4 Non-Functional Requirements:

- 1.) The system will be screen based.
- 2.) Online help will be provided in case of returns/refunds, questions or concerns.
- 3.) CWBS will be a secure, password protected, HTTP website*
- 4.) Clothes can be sorted according to color, size or categories.
- 5.) Multi Language support will be available to cater to all.

2.5 Other Important Assumptions:

- 1.) CWBS can be run on any web browser like firefox, chrome or safari
- 2.) Passwords will need to be changed every eight months to maintain safety of personal information.
- 3.) A website builder will be used for the basic foundation of CWBS.

2.6 Examples of System Input/Output:

The platform that will be used for this is a website, so all the input would be information needed for the website like description of clothing and pictures of models styling the clothes. The output would be invoices of the payments made by customers.

2.7 Knowledge Acquisition:

Initially, we will develop our requirements based on prior knowledge about online clothing stores and how we interact with existing ones. We would also do research and gather information from previous clothing stores. From there, we would analyze the information they have and use it as a reference to build our website.

2.8 Software and/or hardware involved:

The website is browser-based and can be accessed through a PC running on Windows 8 or newer or on Mac's latest version. Recommended browser software for the website is Firefox, Chrome, Internet Explorer or Safari. A mobile phone can also be used to access the website using the browsers mentioned above. We will be using the Lucid chart website and Microsoft Visio for UML Diagrams and system designs.

2.9 Proposed Deliverables and Work Plans:

We will be turning in the following documents as the final project submission with a set of UML diagrams:

- The Problem Statement
- The Requirements Document (Stories)
- The final Use Case Diagram and its explanation
- Use Case Descriptions
- The final class diagram (Domain Model) and its explanation
- System Sequence Diagrams
- Sequence Diagrams
- Design Class Diagram and its explanation

Then our final report will include individual lessons we have learned, how we divided work equally, and mention of any unsolved problems we were not able to complete or achieve in this project.

(3) Use Case Model:

3.1 Requirements Document “Stories”:

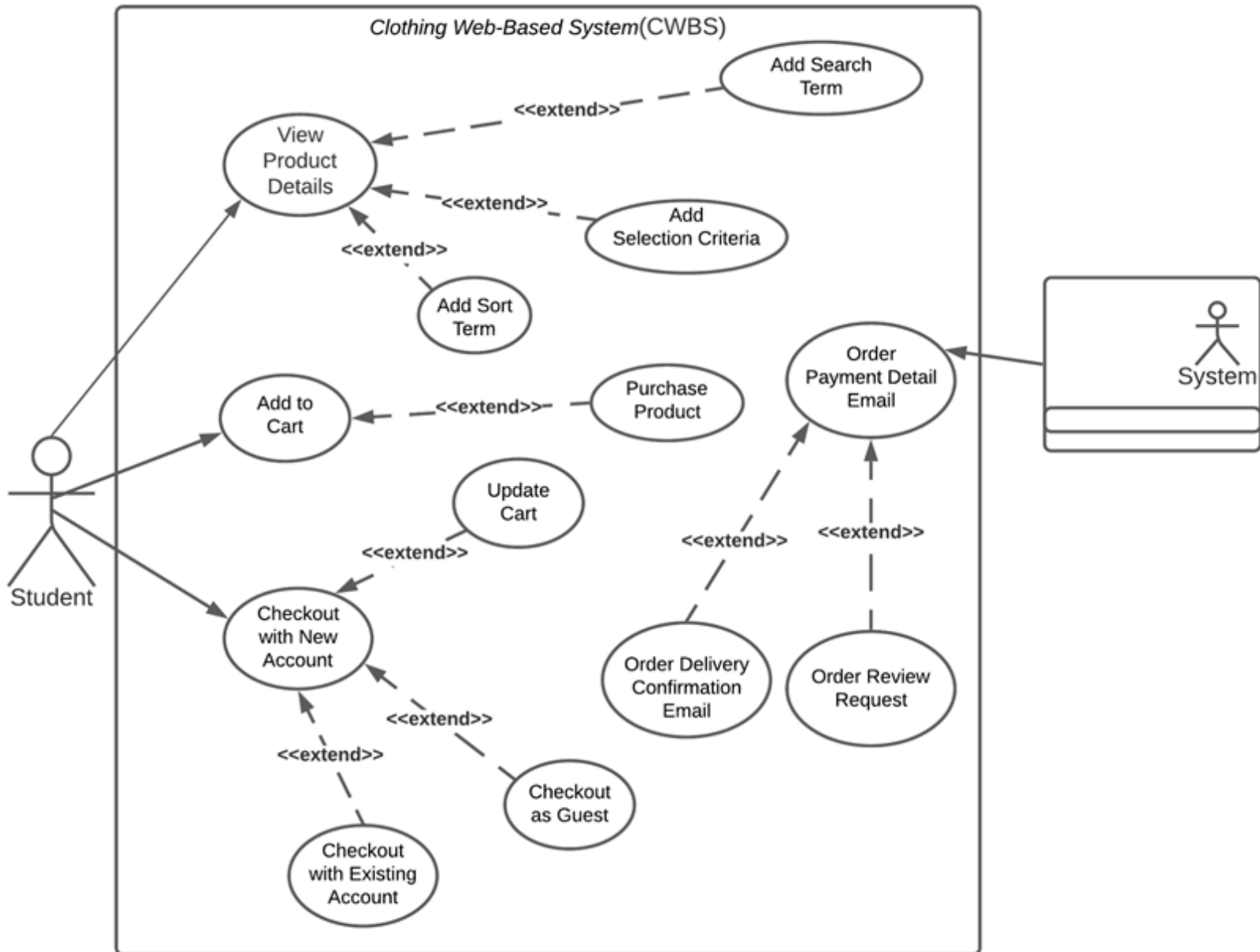
After getting accepted for a job through the Drexel University CO-OP program, a student looks up our new clothing website to shop for some work clothes, as the company only allows corporate clothes. There are different clothing options on our website for the student to look through and choose a particular category they are looking for. After selecting the category that suits them, the system shows products related to that category. The student can make their search easier by sorting the products according to color, size, and type. The system then displays the sorted product list according to the student's preference. If the student wants a product, they can select it and the system then display a page with product details. The student can go over product details and reviews once satisfied with the selection, the student can add the product to cart. The system gives an option of updating color, size, and type within the cart and displays the option of creating a new account or checkout as a guest. The system collects and stores data only when the student chooses to make an account and collects a payment based on product price and shipping cost. Once an order is placed, the system sends out order and payment details to the student email with the option to receive text updates about their order.

A few days after the student receives their order, the system sends out an email confirmation of the order's arrival with a link to review the product with an option to upload a picture of the product. The student can also make a complaint in case they don't receive their order. After reviewing the product, the system updates the order confirmation page and product review section with the latest review on top. The student has a maximum of 30 days to return their products for a refund if they are not satisfied.

3.2 Actors and their Goals:

Student: The main goal of the student is to find and purchase work clothes for a job.

3.3 Use Case Diagram (Entire Project):



3.4 Overview section of all the use cases:

Use Cases	Includes/Extends	Description
View Product Details	Add search term <<Extended>> Add selection criteria <<Extended>> Add sort term <<Extended>>	The student can add the search to view product details or simply select any of the available criteria and view product details. The product details page can also be viewed by adding sort terms such as size, color, etc.
Add to Cart	Purchase Product(s) <<Extended>>	The student can add the product to cart after viewing the product details page. Then once decided about the purchase can proceed towards the purchase product step.
Checkout with new account	Update Cart <<Extended>> Checkout as Guest <<Extended>> Checkout w/existing account <<Extended>>	The student can check out with the new account or just login into an existing account and proceed towards payment. The student also has the option of checking out as a guest. Option to update the cart before the payment can also be used by the student in case they change their mind to not purchase it or purchase something else.
Order Payment Detail Email	Order Confirmation Email <<Extended>> Order Review Request <<Extended>>	Once an order is placed the system orders an email with payment details. Then the system orders an email for delivery confirmation the day order is delivered. Finally the system orders a review request from the student.
Add Search Term	None	Student can add search terms to find the product desired or looking for by putting keywords in the search bar.
Add Selection Criteria	None	Student can select different criteria options to get the product results of that specific criteria.
Add Sort Term	None	Student can sort the product based on the size, color, etc.

Purchase Product	None	Once decided about the product, students can purchase the product preceding the checkout.
Update Cart	None	Products in the cart can be updated such as the quantity of the product or can be removed from the cart to add another product.
Checkout as Guest	None	Student can checkout as guests without having an account to purchase the product.
Checkout with Existing Account	None	Student can sign in to their existing account to purchase the product.
Order Delivery Confirmation Email	None	Once an order is placed and is delivered, the system sends out an email to confirm the delivery of the package.
Order Review Request	None	System sends another email to get a review about the product.

3.5 Use case descriptions:

<i>USE CASE #</i>	VPD01
<i>USE CASE Name</i>	View Product Details
<i>Scenario</i>	The student has selected a product to view that they are interested in
<i>Trigger</i>	This event is triggered when students click on “View Product Details”
<i>ACTOR</i>	Student
<i>Goal (1 phrase)</i>	To view product details
<i>Overview and scope</i>	The student is shopping for professional attire and is trying to view the details of each product before making a purchase
<i>Level</i>	Primary
<i>Stakeholders</i>	Student, the system
<i>Preconditions</i>	Website has to go live
<i>Postconditions in words (write in passive and past tense)</i>	The CWBS system will have the information of students that decide to make an account saved, along with their payment details as well.

Included Use Cases	none	
Extending Use Cases	Add search term, add selection criteria, add sort term	
MAIN SUCCESSFUL SCENARIO in numbered sequence Reference “included use cases” in this section using INCLUDE <i>ius_name</i> Italicized terms are defined in Glossary	Actor Action	System Action
	1. Student logs into website	2. System verifies login
	3. Student browses product according to sort term	4. System displays product based on the sort term
	5. Student views selected product and decides to add to cart	7. System adds the product to cart
OTHER SUCCESSFUL SCENARIOS (Specify any <i>successful</i> variations of the <i>normal</i> execution path, including any extension points using EXTEND <i>eus_name</i>)	Step	Branching Action
	1. Student logs into website	2. System verifies login
	3. Student browses product according to sort term	4. System is unable to display the products based on the sort term
	5. Student views products that are not sorted	
UNSUCCESSFUL SCENARIOS (<i>erroneous</i> situations)	Conditions	Actions
	1. Student views products that are no longer available on the website	2. System does not update the website accordingly to the available products
Priority in scheduling	Primary	
Frequency	Everytime a student wants to view a product	
Business rules and data logic	A product can be viewed when logged on the website You can only view the details of one product at a time on a page	
Other non-functional requirements	none	
Superordinates	N/A	
Developer	Aisha Ali-Ahmad	
Creation date and last modified date	08/05/2021 08/07/2021	
Glossary	N/A	
Other Comments	N/A	

USE CASE #	ATC02	
USE CASE Name	Add to Cart	
Scenario	The student likes something from the products page and decides to purchase it but wants to look at other products. The student simply adds it to the cart and looks for other products or purchases that product through checkout	
Trigger	When the student clicks the “Add to Cart” button	
ACTOR	Student	
Goal (1 phrase)	To add the product to the cart that the student like to purchase	
Overview and scope	This process allows the student to add clothes from the product page they like and need for the job. To add clothes to the cart, students are not required to make an account. The student can look for the products and simply add them to the cart	
Level	Primary	
Stakeholders	Students	
Preconditions	The student needs a stable connection to the internet and access to a device such as mobile or pc to access the website online. Then needs to find a product to add to the cart	
Postconditions in words (write in <i>passive and past tense</i>)	A new product object has been created with the add to cart object. Value of new product object has been added to the add to cart object which has changed the value(s) of the cart Both objects have been deleted Product(s) have been added.	
Included Use Cases	none	
Extending Use Cases	Purchase Product	
MAIN SUCCESSFUL SCENARIO in numbered sequence Reference “included use cases” in this section using INCLUDE <i>ius_name</i> Italicized terms are defined in Glossary	Actor Action	System Action
	1. Student add the search term on the website for clothes	2. System look for the search term and display products
	3. Student scroll down to look at different clothes	4. System show available clothes
	5. Student like the clothes and click on it	6. System display the clothes detail page
	7. Student add it to the cart	8. System update the cart with the new product
	9. Student look at the cart to confirm the product is added and proceed to look at other products	
OTHER SUCCESSFUL SCENARIOS (Specify any <i>successful</i> variations)	Step	Branching Action
	4a Student add selection criteria	5a. System show available clothes based on selection criteria

of the <i>normal</i> execution path, including any extension points using <i>EXTEND eus name</i>)	4b Student add sort-term	5b. System show available clothes based on short-term
UNSUCCESSFUL SCENARIOS (<i>erroneous</i> situations)	Conditions	Actions
	1a. Search term not matched to the available product	2a. System returns with the message no such product was found
	7a. Student add out of stock clothes	8a. The system ignores the command
Priority in scheduling	First	
Frequency	Every time the student wants to add the product to the cart	
Business rules and data logic	The product can only be added if available in stock The same product can't be added more than 10 times in a single purchase If the number of available products is less than 10 then only that number of products can be added to the cart	
Other non-functional requirements	none	
Superordinates	none	
Developer	Junaid Asim	
Creation date and last modified date	08/07/2021	
Glossary	none	
Other Comments	none	

USE CASE #	CNA03
USE CASE Name	Checkout with New Account
Scenario	A student has set up a new account on our platform
Trigger	The event is triggered once the student finished the "sign up" requirements.
ACTOR	Student
Goal (1 phrase)	To access CWBS personal account for convenience of saving information and tracking previous purchases.
Overview and scope	The student will be able to create an account or check out as a guest. If an account is chosen, he/she can sign up and sign in later on.

Level	Primary	
Stakeholders	Student and the system	
Preconditions	Student has to choose to sign up	
Postconditions in words (write in passive and past tense)	A new account was created The new account was stored to system An order was made Once the order was placed, the payment email was sent out	
Included Use Cases	None	
Extending Use Cases	<<Update Cart>> <<Checkout as Guest>> <<Checkout w/existing account >>	
MAIN SUCCESSFUL SCENARIO in numbered sequence Reference “included use cases” in this section using INCLUDE <i>ius_name</i> Italicized terms are defined in Glossary	Actor Action	System Action
	1. Student opens CWBS platform	2. System gives option for log in, sign up or guest
	3. Student clicks on sign up button	4. System open sign up pages with all information needed for sign up
	5. Student fills in all information and clicks “Submit”	6. System verifies students and information provided. System also sends verification email
	7. Student completes verification and signs in	8. System allows for sign in and sign up is complete
OTHER SUCCESSFUL SCENARIOS (Specify any <i>successful</i> variations of the <i>normal</i> execution path, including any extension points using EXTEND <i>eus_name</i>)	Step	Branching Action
	9. Student tries to check out from CWBS system	10 System picks up no previous log in and ask “checkout as guest” or “sign up”
	11 Student clicks on sign up button	12 System open sign up pages with all information needed for sign up
	13 Student fills in all information and clicks “Submit”	14 System verifies students and information provided. System also sends verification email
	15 Student completes verification and signs in	16 System allows for sign in and sign up is complete
UNSUCCESSFUL SCENARIOS (<i>erroneous</i> situations)	Conditions	Actions
	17. Student tries to sign up	18. System sends “are you a robot?” verification

	19. Student does not pick the right answers and hence fails verification	20. System shows failed verification
Priority in scheduling	Primary/Optional	
Frequency	Every time a new student comes to the system and chooses to sign up.	
Business rules and data logic	21. Payment can be in full or made using a four time payment system. 22. Payment can be made using debit or credit card. 23. System cancels order if payment is not received within 10 days of order placement	
Other non-functional requirements	All personal information giving will be encrypted in the system Email and password can also be saved for easy log in in future visits. Password will need to be updated at least every six months	
Superordinates		
Developer	Tobi Bakare	
Creation date and last modified date	08/05/2021 08/07/2021	
Glossary	none	
Other Comments	none	

<i>UE CASE #</i>	OPDE04
<i>USE CASE Name</i>	Order Payment Detail Email
<i>Scenario</i>	When a student places an order, the system will send an email for with payments details and confirmation
<i>Trigger</i>	The event is triggered once an order is placed (“Place order”)
<i>ACTOR</i>	Student
<i>Goal (1 phrase)</i>	To receive order payment details email
<i>Overview and scope</i>	The student has found all professional attires wanted/needed and is about to place order and receive confirmation of order.
<i>Level</i>	Primary
<i>Stakeholders</i>	student and the system
<i>Preconditions</i>	Students has clicked “Place order” button

Postconditions in words (write in passive and past tense)	“NewOrder” object was created and associated with the following attributes: Order.Number”: to identify each order made Order.Email”: will be sent out once order number is created Student.Details”: email and shipping details of student	
Included Use Cases	None	
Extending Use Cases	<<Order Delivery Confirmation Email>> <<Order Review Request>>	
MAIN SUCCESSFUL SCENARIO in numbered sequence Reference “included use cases” in this section using INCLUDE <i>ius_name</i> Italicized terms are defined in Glossary	Actor Action	System Action
	1. Student click on the “Place Order” button.	2. System allows for orders to be placed and sends confirmation email with payment and shipping details.
	3. Student click on email or link in email.	4. The system provides all information about order needed by the student.
OTHER SUCCESSFUL SCENARIOS (Specify any <i>successful</i> variations of the <i>normal</i> execution path, including any extension points using EXTEND <i>eus_name</i>)	Step	Branching Action
	5. Student puts in a phone number for confirmation details instead of email.	6. System allows for orders to be placed but sends confirmation text with payment and shipping details.
	7a Student pays by cash	8a. System indicates cash and enter amount
	7b Student pays by check	8b. System indicates check and enter amount
UNSUCCESSFUL SCENARIOS (<i>erroneous</i> situations)	Conditions	Actions
	9. Student click on the “Place Order” button.	10. System allows for orders to be placed but doesn’t send confirmation emails with payment and shipping details.
	11. Student is short of money;	12. Abort the transaction
	13 Credit card validation fails	14. Ask to pay by another means or abort
Priority in scheduling	Primary	
Frequency	Every time an order is placed	

Business rules and data logic	Payment can be in full or made using a four time payment system. Payment can be made using debit or credit card. System cancels order if payment is not received within 10 days of order placement
Other non-functional requirements	Payment details and address will be encrypted in the system Order tracking will be updated constantly
Superordinates	
Developer	Tobi Bakare
Creation date and last modified date	08/05/2021 08/07/2021
Glossary	none
Other Comments	none

<i>USE CASE #</i>	CAG02	
<i>USE CASE Name</i>	Checkout as Guest	
<i>Scenario</i>	When a student does not want to create an account but wants to make a purchase	
<i>Trigger</i>	The event is triggered once “Checkout as Guest” is clicked	
<i>ACTOR</i>	Student	
<i>Goal (1 phrase)</i>	To make an order without an existing or new account	
<i>Overview and scope</i>	The student has found all professional attires wanted/needed and is about to checkout for the first time but does not want to go through the process of inputting account details	
<i>Level</i>	<<Extended>>	
<i>Stakeholders</i>	student	
<i>Preconditions</i>	Students has clicked “Checkout as Guest” button	
<i>Postconditions in words (write in passive and past tense)</i>	“GuestCheckout” object has been created with the updates in the following attributes: GuestInformation: will be collected because it is not saved in the system GuestCheckout object has been destroyed	
<i>Included Use Cases</i>	none	
<i>Extending Use Cases</i>	none	
<i>MAIN SUCCESSFUL SCENARIO in numbered sequence</i>	<i>Actor Action</i>	<i>System Action</i>
	Student click on the “Place Order” button.	2. System allows for orders to be placed and sends confirmation email with payment and shipping details.

Reference “included use cases” in this section using INCLUDE <i>ius_name</i>	3. Student click on email or link in email.	4. The system provides all information about order needed by the student.
Italicized terms are defined in Glossary		
OTHER SUCCESSFUL SCENARIOS (Specify any <i>successful</i> variations of the <i>normal</i> execution path, including any extension points using EXTEND <i>eus_name</i>)	Step	Branching Action
	1a. Student click on the “Check out” button.	2a. System automatically takes you through check out and to put in necessary information
	3a. Student click checkout as guest after filling in information	4a. System allows for order to be made after clicking the check out as a guest button.
UNSUCCESSFUL SCENARIOS (<i>erroneous</i> situations)	Conditions	Actions
	1. Student click on the “Check out as a Guest” button.	2. System does not allow for it because student is logged in and has existing account
Priority in scheduling	Optional	
Frequency	Every time a new student without an account wants to checkout	
Business rules and data logic	1. Payment can be in full or made using a four time payment system. 2. Payment can be made using debit or credit card. 3. System cancels order if payment is not received within 10 days of order placement 4.	
Other non-functional requirements	Students can checkout as a guest as many times as they want	
Superordinates		
Developer	Tobi Bakare	
Creation date and last modified date	08/26/2021 08/26/2021	
Glossary	None	
Other Comments	None	

USE CASE #	AST02
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USE CASE Name	Add Search Term	
Scenario	A student can search the items keyword through a specific block of the web application for better convenience	
Trigger	Student will click on the search bar after the login to view products	
ACTOR	Student	
Goal (1 phrase)	To search for the various products available on the website and to save their time	
Overview and scope	Students will be able to search the products after just doing login with their login credentials and can find the products of their choice without browsing the website or as guest also	
Level	<<Extended>>	
Stakeholders	Student	
Preconditions	Student has entered the website and enter the valid product name	
Postconditions in words (write in passive and past tense)	“searchTerm” have been created with the updates in the following attributes: searchByName: will search the product by its name searchById: will search the product by its id “SearchTerm” have been destroyed	
Included Use Cases	None	
Extending Use Cases	None	
MAIN SUCCESSFUL SCENARIO in numbered sequence Reference “included use cases” in this section using INCLUDE <i>ius_name</i> Italicized terms are defined in Glossary	Actor Action	System Action
	1. Student search for the product	2. system search for the products in the website database
	3. Student clicks on product	4. System provides the products details according to the student search keyword
	5a. student can add the product to the cart or may proceed for the checkout	6. System verifies the student and the product selected.
OTHER SUCCESSFUL SCENARIOS (Specify any <i>successful</i> variations of the <i>normal</i> execution path, including any extension points using EXTEND <i>eus_name</i>)	Step	Branching Action
	5b. student can directly search for the item without login	6a. system finds the provided product name

UNSUCCESSFUL SCENARIOS (erroneous situations)	Conditions	Actions
	1.student enters an invalid product name	1.system cannot find the provided product name
Priority in scheduling	Primary/Optional	
Frequency	Every time a student will search the products	
Business rules and data logic	1. Student can fill in the full product name 2. System will suggest the name as the student tries to search the products.	
Other non-functional requirements	Student search history could be saved for the same future activities, will save the time of the student	
Superordinates		
Developer	Junaid Asim	
Creation date and last modified date	08/15/2021 08/29/2021	
Glossary	None	
Other Comments	None	

USE CASE #	ASC03
USE CASE Name	Add Selection Criteria
Scenario	A student can guide the system how the search should be like according to his selection criteria
Trigger	Student will be asked to set the filters for a better search of the products
ACTOR	Student
Goal (1 phrase)	To provide the products to the student as per his need with various filters so that student will be comfortable in all the ways
Overview and scope	The student will be able to get the products according to him by just adding his selection criteria
Level	<<Extended>>
Stakeholders	Student and the system
Preconditions	Student have to add the selection criteria
Postconditions in words (write in passive and past tense)	“selectionCriteria” object have been created Updates in attributes are below: byprice: will show the items with the selected price range byCategory: will show the items by category selected byName: will show the product its name “selectionCriteria” have been destroyed
Included Use Cases	None

Extending Use Cases	None	
MAIN SUCCESSFUL SCENARIO in numbered sequence Reference “included use cases” in this section using INCLUDE <i>ius_name</i> Italicized terms are defined in Glossary	Actor Action	System Action
	1. Student enters in the website	2. System gives option to search the products by student’s requirements
	3. Student adds all the selections criteria to find the appropriate product	4. System finds the product and will display its details
OTHER SUCCESSFUL SCENARIOS (Specify any <i>successful</i> variations of the <i>normal</i> execution path, including any extension points using EXTEND <i>eus_name</i>)	Step	Branching Action
	3a. students searches by only using single attribute	4a. System picks up appropriate product
UNSUCCESSFUL SCENARIOS (<i>erroneous</i> situations)	Conditions	Actions
	4b. Student doesn’t select any selection criteria	5. system won’t show the products
Priority in scheduling	Primary/Optional	
Frequency	Every time an existing student comes search the product	
Business rules and data logic	1. Student can go for this multiple times 2. It’s not mandatory for the student to use this selection criteria	
Other non-functional requirements	Students all personal information should be encrypted Website should support by all browsers	
Superordinates	None	
Developer	Junaid Asim	
Creation date and last modified date	08/15/2021 08/29/2021	
Glossary	None	
Other Comments	None	

USE CASE #	AST12
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USE CASE Name	Add Sort Term	
Scenario	Add Sort option to sort the items in the page	
Trigger	Customers click on to the Sort button and can choose how to sort the items in the page	
ACTOR	Student	
Goal (1 phrase)	To Sort the products from a list of products in a particular order as per the customer, for example sort by rating, sort by price etc.	
Overview and scope	This option allows customers to access relevant data from a large data set and provide the user the list of products as per his convenience and feasibility	
Level	<<Extended>>	
Stakeholders	Students	
Preconditions	- Customers must visited to the website - Customers must search for the objects in order to sort the list of products	
Postconditions in words (write in passive and past tense)	“Sort” object has been created The following attributes have been updated: Sort “Created” Price: will sort the list by price Rating: will sort the list the ratings in a particular order(ascending or descending) mostRelevant: will sort the items which are the most relevant object has been destroyed	
Included Use Cases	None	
Extending Use Cases	None	
MAIN SUCCESSFUL SCENARIO in numbered sequence Reference “included use cases” in this section using INCLUDE <i>ius_name</i> Italicized terms are defined in Glossary	Actor Action	System Action
	1. Customers visit the website and choose “Sort”	2. System will generate a query to sort and ask user for the type
	3. Students selects the appropriate option for sorting	4. system will evaluate the products as per sorting requirements
	5. Customers can view the modified results and continue further	6. System will display the modified
OTHER SUCCESSFUL SCENARIOS (Specify any <i>successful</i> variations of the <i>normal</i> execution path,	Step	Branching Action
	5a. Customers can change the type	6a. System will display the modified
	5b. Customers now can, browse/access only those items	6b. System will show only required items

including any extension points using EXTEND <i>eus name</i>)		
UNSUCCESSFUL SCENARIOS (<i>erroneous</i> situations)	Conditions	Actions
	7. Customer may not search the products	8. System is an automaton, it will display as per the selection , irrespective of right or wrong selections
Priority in scheduling	Primary	
Frequency	Every time student selects to sort	
Business rules and data logic	- Customer can select sort type - Customer may or may not use filters - Sort will be consisting of multiple type	
Other non-functional requirements	1. Website should be compatible for all browsers 2. Payment information should be encrypted	
Superordinates	None	
Developer	Junaid Asim	
Creation date and last modified date	08/15/2021 08/29/2021	
Glossary	None	
Other Comments	None	

USE CASE #	PPR04
USE CASE Name	Purchase Product
Scenario	After selecting the product(s) student can proceed further to purchase the particular product(s)
Trigger	Student will move to the next page in order to purchase the product
ACTOR	Student
Goal (1 phrase)	To purchase the product that user have selected and completed the order by check out
Overview and scope	This process allows the student to complete the order of purchasing the product
Level	<<Extended>>
Stakeholders	Students
Preconditions	Student need to select at least one of the products from the website in order to purchase the product
Postconditions in words (write in	A new product object has been created called “purchaseProduct” Following attributes have been updated: orderedDate: date when order was placed

<i>passive and past tense)</i>	orderId: id number of the purchased product productPrice: price of the product “purchaseProduct” have been destroyed	
Included Use Cases	None	
Extending Use Cases	None	
MAIN SUCCESSFUL SCENARIO in numbered sequence Reference “included use cases” in this section using INCLUDE <i>ius_name</i> Italicized terms are defined in Glossary	Actor Action	System Action
	1. Student selects the product and view the details	2. System let the product to display to the user
	3. Student proceed to purchase the product	4. a query generated in the system in order to purchase the product
OTHER SUCCESSFUL SCENARIOS (Specify any successful variations of the <i>normal</i> execution path, including any extension points using EXTEND <i>eus_name</i>)	Step	Branching Action
	3a. Student add more items to purchase	4a. System allows to add more items and generate another query
UNSUCCESSFUL SCENARIOS (<i>erroneous</i> situations)	Conditions	Actions
	4. Student haven’t selected any product and proceed for purchase	5. System returns with the message no such product was selected
	5. Student add out of stock clothes	6. The system ignores the command
Priority in scheduling	Primary	
Frequency	Every time the student proceeds to purchase the product.	
Business rules and data logic	- The product can only be added if available in stock - The same product can’t be added more than 10 times in a single purchase - If the number of available products is less than 10 then only that number of products can be added to the cart	
Other non-functional requirements	None	
Superordinates	None	
Developer	Junaid Asim	

Creation date and last modified date	08/15/2021 08/29/2021
Glossary	None
Other Comments	None

USE CASE #	UCT05	
USE CASE Name	Update Cart	
Scenario	The student likes something else from the products page and decides to purchase or want some changes in the cart. It can be added be added to the cart	
Trigger	When the student clicks the “Update Cart” button	
ACTOR	Student	
Goal (1 phrase)	To add the product to the cart that the student like to purchase or wants to update the cart (like change quantity)	
Overview and scope	It will allow the student to add clothes from the product page they like and need for the job. To add clothes to the cart, student is not required to make an account. The student can look for the products and simply add them to the cart	
Level	<<Extended>>	
Stakeholders	Students	
Preconditions	Student need a stable connection to the internet and access to a device such as mobile or pc to access the website online. Then needs to find a product to add to the cart	
Postconditions in words (write in passive and past tense)	A new product object has been created with the “updateCart” Following attributes have been updated addMoreitems: able to add more items changeProduct: able to cancel selected product or replace it changeQuantity: able to change quantity “updateCart” has been destroyed	
Included Use Cases	None	
Extending Use Cases	None	
MAIN SUCCESSFUL SCENARIO in numbered sequence Reference “included use cases” in this section using INCLUDE <i>ius_name</i>	Actor Action	System Action
	1. Student add the items in the cart	2. System look for the search term and add it into cart
	3. Student changes the quantity of the product	4. System updates the quantity
	5. Student replace the product with another one	6. System display the clothes detail page
	7. Student add it to the cart	8. System update the cart with the new product

Italicized terms are defined in Glossary	9. Student look at the cart to confirm the product is added and proceed to look at other products	
OTHER SUCCESSFUL SCENARIOS (Specify any <i>successful</i> variations of the <i>normal</i> execution path, including any extension points using EXTEND <i>eus name</i>)	Step	Branching Action
	5a. Student add selection criteria	6a. System show available clothes based on selection criteria
	5b. Student add sort-term	6b. System show available clothes based on sort-term
UNSUCCESSFUL SCENARIOS (<i>erroneous</i> situations)	Conditions	Actions
	Search term not matched to the available product	System returns with the message no such product was found
	Student add out of stock clothes	The system ignores the command
Priority in scheduling	First	
Frequency	Every time the student wants to add more product or wants any changes in the cart	
Business rules and data logic	- The product can only be added if available in stock - The same product can't be added more than 10 times in a single purchase - If the number of available products is less than 10 then only that number of products can be added to the cart	
Other non-functional requirements	None	
Superordinates	None	
Developer	Junaid Asim	
Creation date and last modified date	08/15/2021 08/29/2021	
Glossary	None	
Other Comments	None	

USE CASE #	ASC09
USE CASE Name	Checkout with Existing Account
Scenario	Student must continue with their existing account after just filling their login credentials

Trigger	The event is triggered once the student finished the “sign up” requirements	
ACTOR	Student	
Goal (1 phrase)	To access CWBS personal account for convenience of saving information and tracking previous purchases	
Overview and scope	The student will be able to continue with their existing account. All their previous activities, cart list, saved items etc, will remain same	
Level	<<Extended>>	
Stakeholders	Student and the system	
Preconditions	Student must fill login credentials	
Postconditions in words (write in passive and past tense)	Student do login An order was made Once the order was placed, the payment email was sent out	
Included Use Cases	None	
Extending Use Cases	None	
MAIN SUCCESSFUL SCENARIO in numbered sequence Reference “included use cases” in this section using INCLUDE <i>ius_name</i> Italicized terms are defined in Glossary	Actor Action	System Action
	1. Student opens CWBS platform	2. System gives option for log in, sign up or guest
	3. Student clicks on sign up button	4. System open sign-up pages with all information needed for sign up
	5a. Student fills in all information and clicks “Submit”	6. System verifies students and information provided. System also sends verification email
	7. Student completes verification and signs in	8. System allows for sign in and sign up is complete
OTHER SUCCESSFUL SCENARIOS (Specify any <i>successful</i> variations of the <i>normal</i> execution path, including any extension points using EXTEND <i>eus_name</i>)	Step	Branching Action
	9b. Student tries to login as an existing user and fill login info	10a. System picks up login with saved login information
	9b. Student completes with the login info and signs in	10b. System allows for sign in
UNSUCCESSFUL SCENARIOS (<i>erroneous</i> situations)	Conditions	Actions
	9c. Student tries to sign up	10c. System sends “are you a robot?” verification

	9d. Student does not fill the correct login info	10d. System shows invalid login
Priority in scheduling	Primary/Optional	
Frequency	Every time an existing student comes to the system and chooses to sign up	
Business rules and data logic	- Payment can be in full or made using a four-time payment system - Payment can be made using debit or credit card - System cancels order if payment is not received within 10 days of order placement	
Other non-functional requirements	Previous login information could be saved and all personal information given will be encrypted in the system. Password will need to be updated at least every six months	
Superordinates	None	
Developer	Junaid Asim	
Creation date and last modified date	08/15/2021 08/29/2021	
Glossary	None	
Other Comments	None	

<i>USE CASE #</i>	ODCE07
<i>USE CASE Name</i>	Order Delivery Confirmation Email
<i>Scenario</i>	When the student will proceed further for payment, he can take the confirmation the order delivery through email
<i>Trigger</i>	When the student chooses to get the order delivery confirmation through email
<i>ACTOR</i>	Student
<i>Goal (1 phrase)</i>	To get the confirmation email about the order delivery for the product(s) student have selected
<i>Overview and scope</i>	This process allows the student to get the email confirming for the order deliver
<i>Level</i>	<<Extended>>
<i>Stakeholders</i>	Students
<i>Preconditions</i>	Student needs a to make a order purchase with at least one item in their cart
<i>Postconditions in words (write in</i>	A new object “orderDelConf” have been created Following attributes have been updated: email: email address of the student

<i>passive and past tense)</i>	confirm: attribute with the bool data type “orderDelConf” have been destroyed	
Included Use Cases	None	
Extending Use Cases	None	
MAIN SUCCESSFUL SCENARIO in numbered sequence Reference “included use cases” in this section using INCLUDE <i>ius_name</i> Italicized terms are defined in Glossary	Actor Action	System Action
	1. Student enters the email	2. System look for the entered email whether it is valid or not
	3. Student make the choice of confirmation in yes or no	4. System accepts the choice
OTHER SUCCESSFUL SCENARIOS (Specify any successful variations of the <i>normal</i> execution path, including any extension points using EXTEND <i>eus_name</i>)	Step	Branching Action
	3a. Student chooses yes	4a. System sent the email for confirmation
	3b. Student chooses No	4b. System do nothing
UNSUCCESSFUL SCENARIOS (<i>erroneous</i> situations)	Conditions	Actions
	Student have not selected any item for placing order	System returns with the message no product was found to place order
	Student add out of stock clothes	The system ignores the command
Priority in scheduling	First	
Frequency	Every time the student selects to make confirmation by email	
Business rules and data logic	- It can only be possible if student have at least one item in the cart - Email address should be valid	
Other non-functional requirements	None	
Superordinates	None	
Developer	Junaid Asim	
Creation date and last modified date	08/15/2021 08/29/2021	

Glossary	None
Other Comments	None

USE CASE #	ORR08	
USE CASE Name	Order Review Request	
Scenario	When the student will proceed further for payment, he can take the review the order	
Trigger	When the student chooses to review the order	
ACTOR	Student	
Goal (1 phrase)	To show the order details to the student before placing the order for example products selected, total price, quantities, delivery address etc	
Overview and scope	This process allows the student to review the order and items they are going to purchase, and they want some changes they can do it before placing the order	
Level	<<Extended>>	
Stakeholders	Students	
Preconditions	Student needs a to make an order purchase with at least one item in their cart	
Postconditions in words (write in <i>passive and past tense</i>)	A new object has been created “reviewOrder” Attributes have been updates as: showProducts: shows all the products in the cart totalAmount: total amount to be paid object have been destroyed	
Included Use Cases	None	
Extending Use Cases	None	
MAIN SUCCESSFUL SCENARIO in numbered sequence Reference “included use cases” in this section using INCLUDE <i>ius_name</i> Italicized terms are defined in Glossary	Actor Action	System Action
	1. Student request to review order	2. System fetches the order details and it to the student
	3. Student select a particular entity to review	4. System allows to select
OTHER SUCCESSFUL	Step	Branching Action

SCENARIOS (Specify any <i>successful</i> variations of the <i>normal</i> execution path, including any extension points using EXTEND <i>eus name</i>)	3a Student choose to review the products	4a. System shows the selected products
	3b Student choose to review amount	4b. System show the total amount to be paid
UNSUCCESSFUL SCENARIOS (<i>erroneous</i> situations)	Conditions	Actions
	5.Student cart is empty	6.System returns with the message please add items in cart
Priority in scheduling	First	
Frequency	Every time the student wants to review order details	
Business rules and data logic	- It can only be possible if student have at least one item in the cart - Products should be in stock	
Other non-functional requirements	None	
Superordinates	None	
Developer	Junaid Asim	
Creation date and last modified date	08/15/2021 08/29/2021	
Glossary	None	
Other Comments	None	

3.6 Activity diagram:

Activity Diagram for Use Case View Product Details (Junaid Asim):

3.7 Discussion:

For the above activity diagram (Junaid):

We can assume the student as a web user who wants to access the website and then want to place the order and in order to do that we present the activity diagram which is a graphical representation of an executed set of procedural system activities. Student can open any web browser in order to access the website. In the second step Student have to enter the website url. After entering the web url student will enter in the website, here student will have the option to access the website without a registered account also.

(4) Class Model:

4.1 Thinking procedure while developing class diagram:

Step 1: Identify the class names:

The very first step I have taken is by knowing the primary objective of a particular class like what it is going to perform and giving the outputs and trying to build each and every class with appropriate attributes and methods.

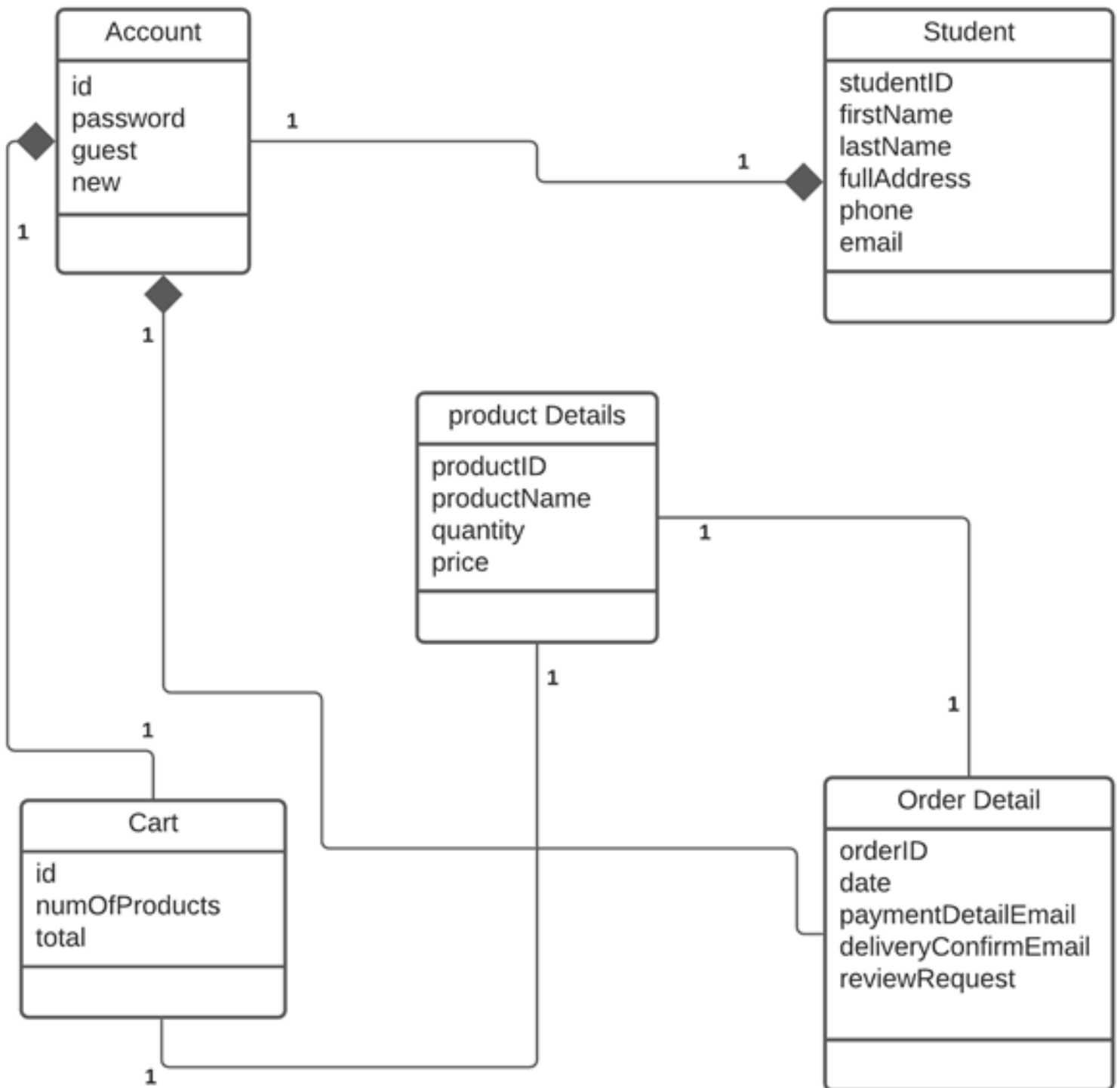
Step 2: Distinguish relationships:

Next step I have taken is to determine how each of the classes or objects are related to one another. Look out for commonalities and abstractions among them, it helps me when I group them when drawing the class diagram.

Step 3: Create the Structure:

First of all I had to add the class names and link them with the appropriate connectors then after I add attributes and functions/methods/operations later.

4.2 Class diagram:



4.3 & 4.4 Selected Class definitions and Association definitions:

- Order Detail: This class have the attributes orderID which will be unique every time the student places an order with the date, paymentDetailEmail, deliveryConfirmEmail and reviewRequest and will interacts to the system directly after getting order from the student to send appropriate emails.
- Account: This class holds all the important information about the user account and plays an important role here because it has a compositional relationship with other 3 classes. It has attributes id which is a unique number for every account, password that will be of the user, guest and new attributes containing Boolean values.
- Cart: This Cart Class allows the website visitors to select, reserve, and purchase a product or service from the interface, it has an attribute id which helps the system to analyze each cart uniquely and numOfProducts attribute which shows the products in the class and total of that products.

4.5 Discussion:

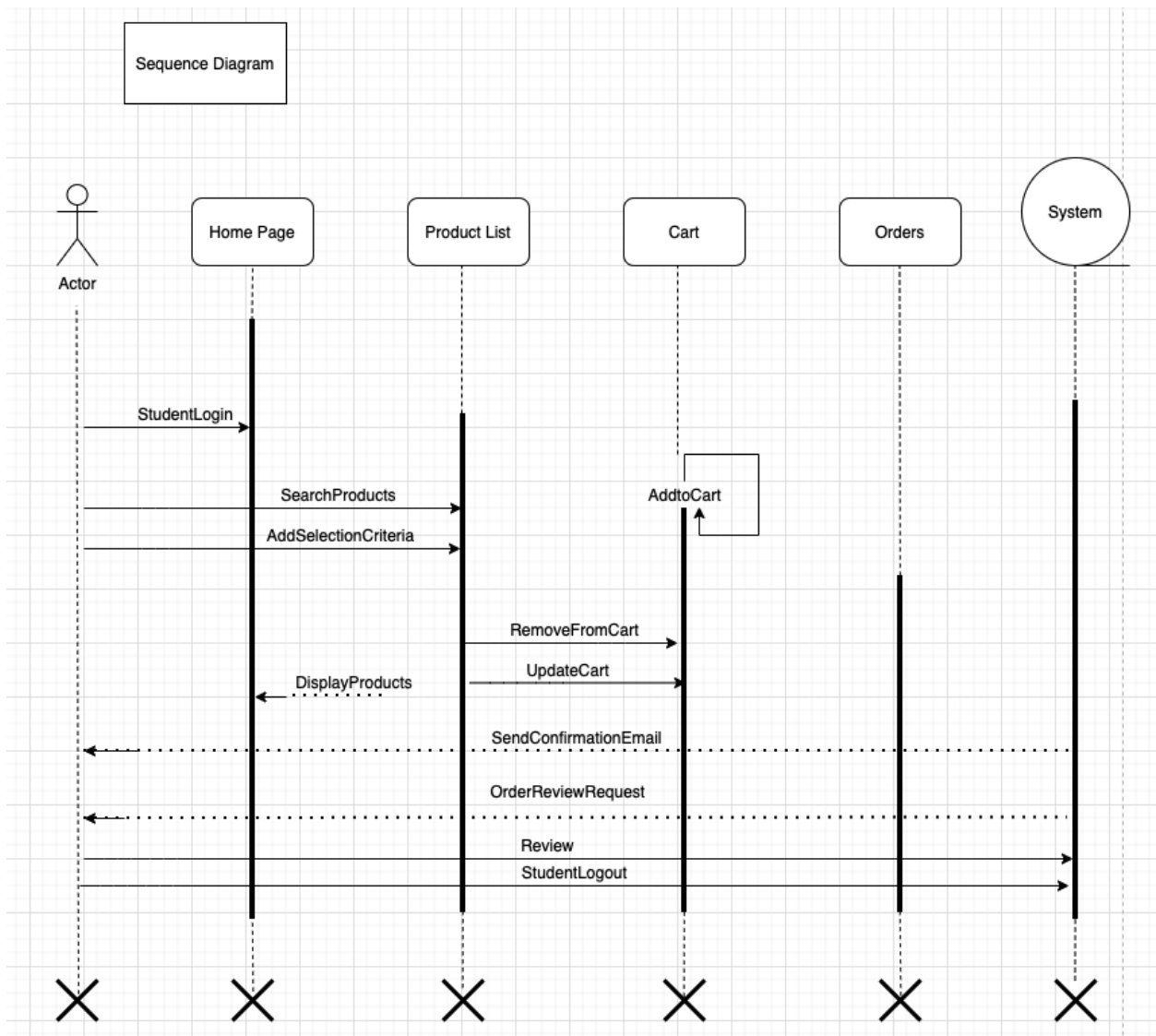
Junaid: Classes are depicted as boxes with three sections, the top one indicates the name of the class, the middle one lists the attributes of the class, and the third one lists the methods. By including both an attribute and a method box in the class I have arguably made design decisions in our model, something I shouldn't be doing if our goal is conceptual modeling. Another approach would be to have two sections, one for the name and one listing responsibilities. I could also use class boxes that show just the name of the class, enabling us to focus on just the classes and their relationships. However, if that was our goal I would be more likely to create a short detailed diagram instead. Account is an associative class, also called a link class, which is used to model associations that have methods and attributes. Associative classes are typically modelled during analysis and then refactored. Attributes and associations are both properties in the UML, so they're treated as basically the same sort of thing. I also showed associations are implemented as a combination of attributes and operations. I prefer to keep our models simple and assume that the attributes and operations exist to implement the associations. Furthermore, that would be a detailed design issue anyway, something that isn't appropriate on a conceptual model.

CHAPTER_2 SYSTEM DESIGN

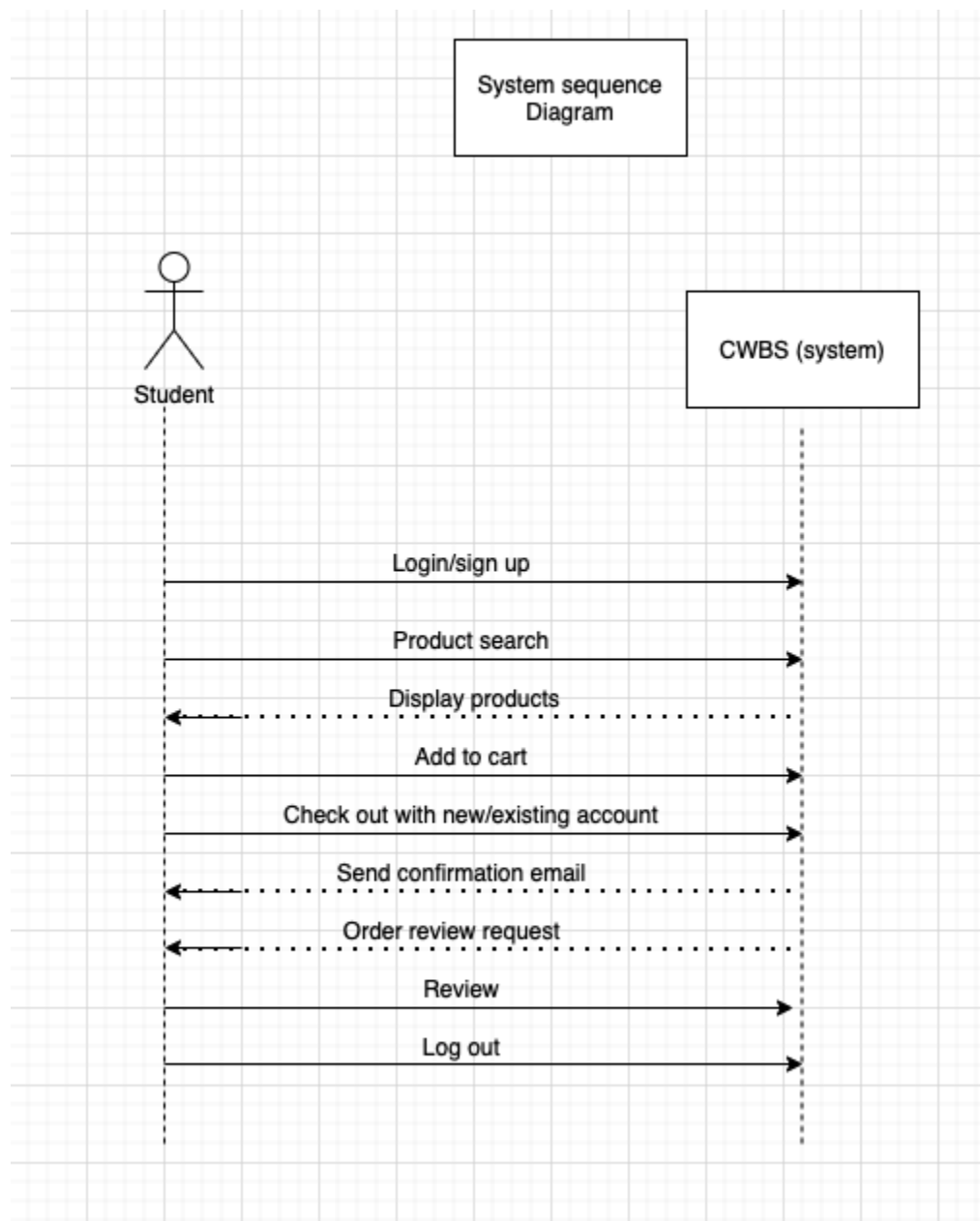
(5) Sequence Diagrams:

CWBS (system)

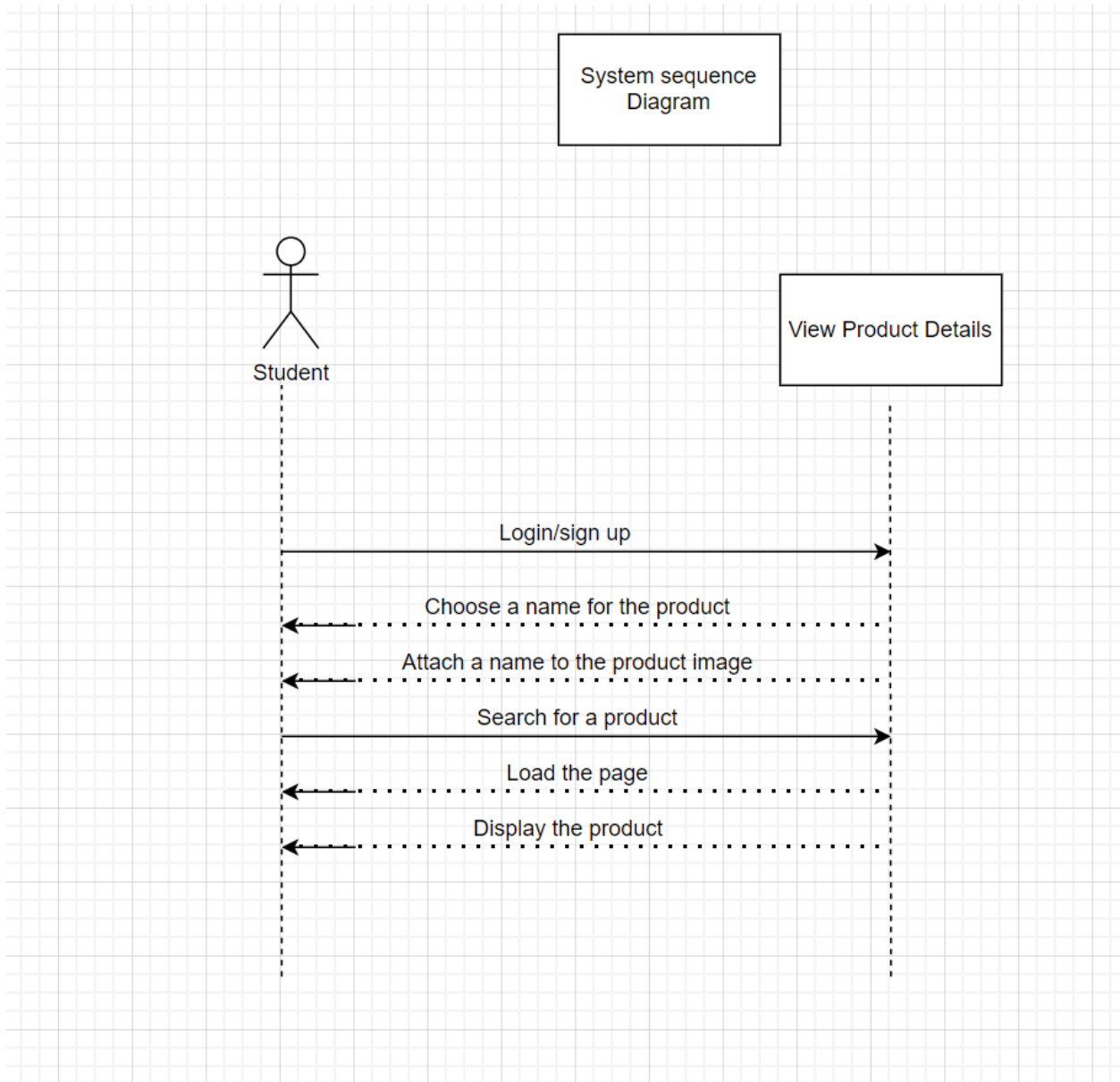
Tobi



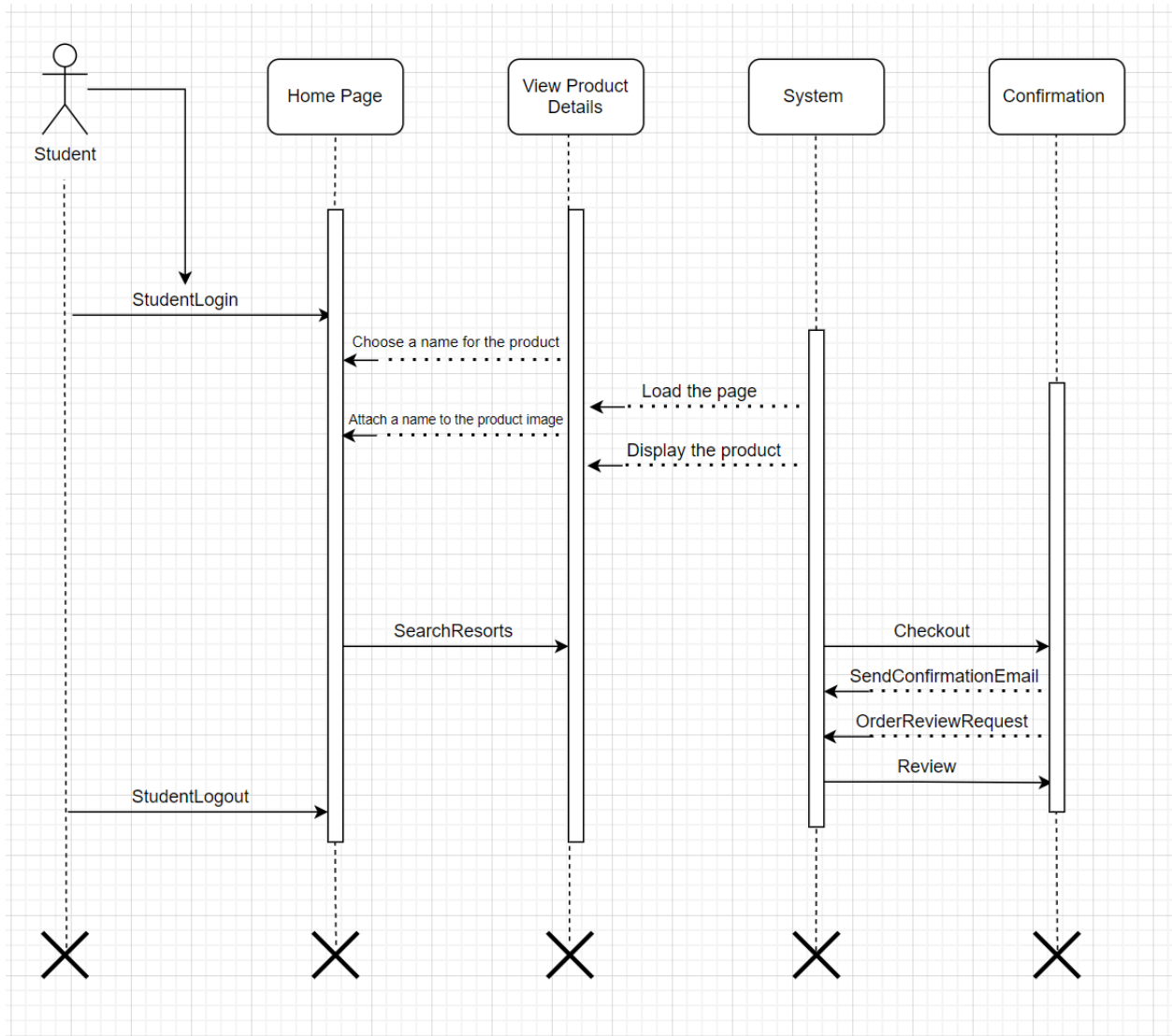
CWBS (system)
Tobi



System Sequence Diagram
View Product Details
Aisha



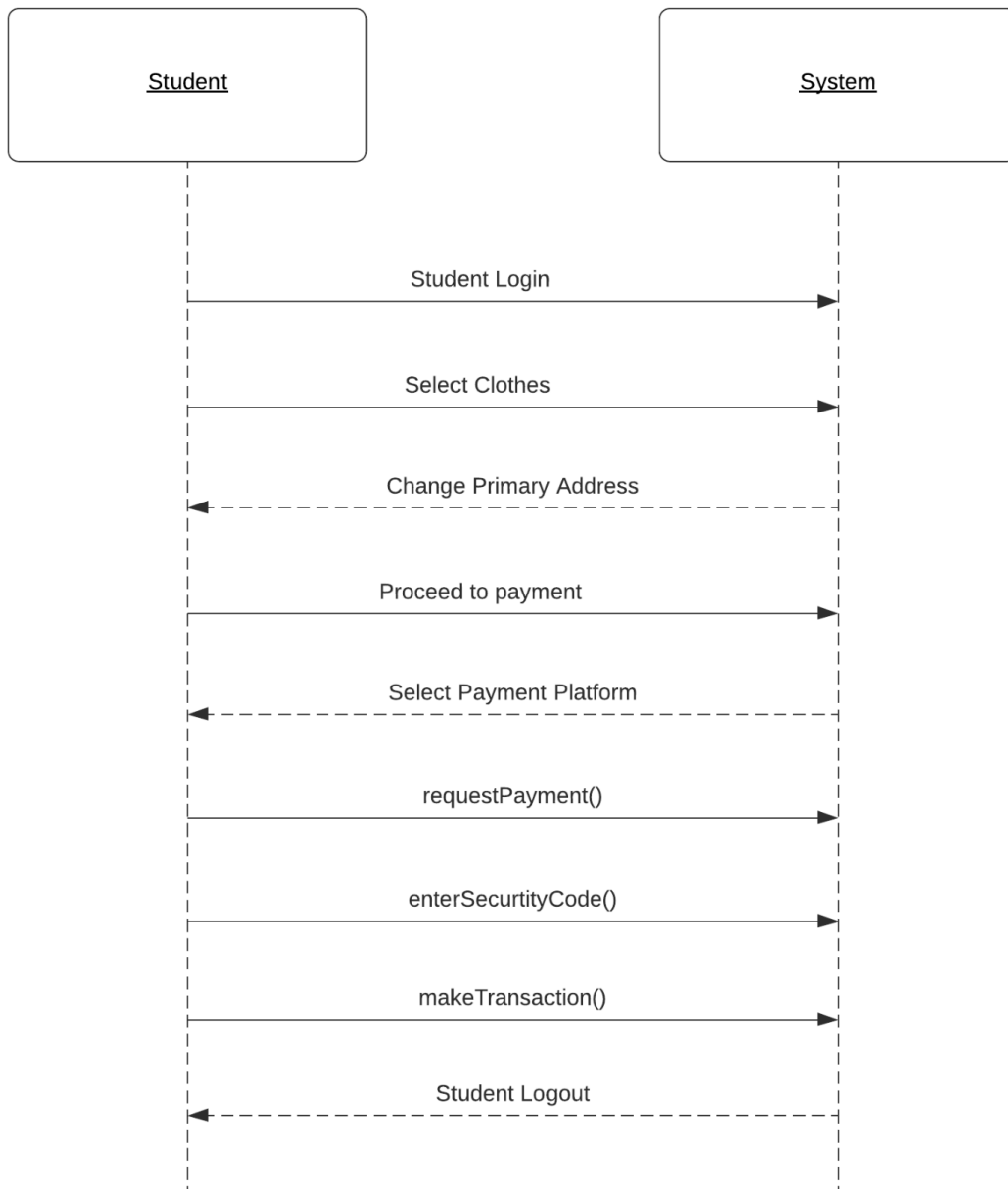
System Sequence Diagram
View Product Details
Aisha



5.1 System sequence diagram:

Junaid

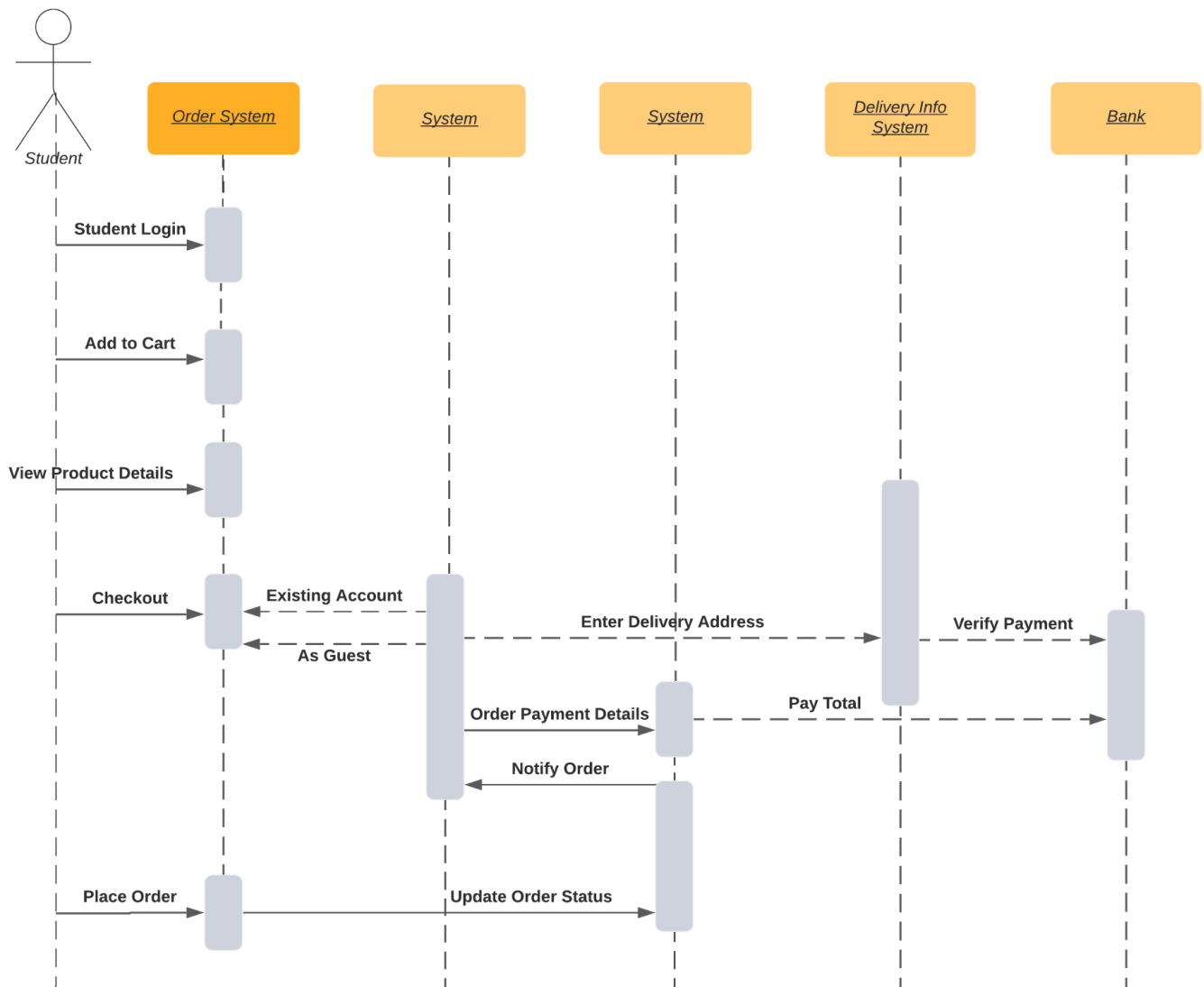
Order Payment Detail Email



5.2 Set of sequence diagrams:

Junaid

Order Payment Detail Email



5.3 Discussion:

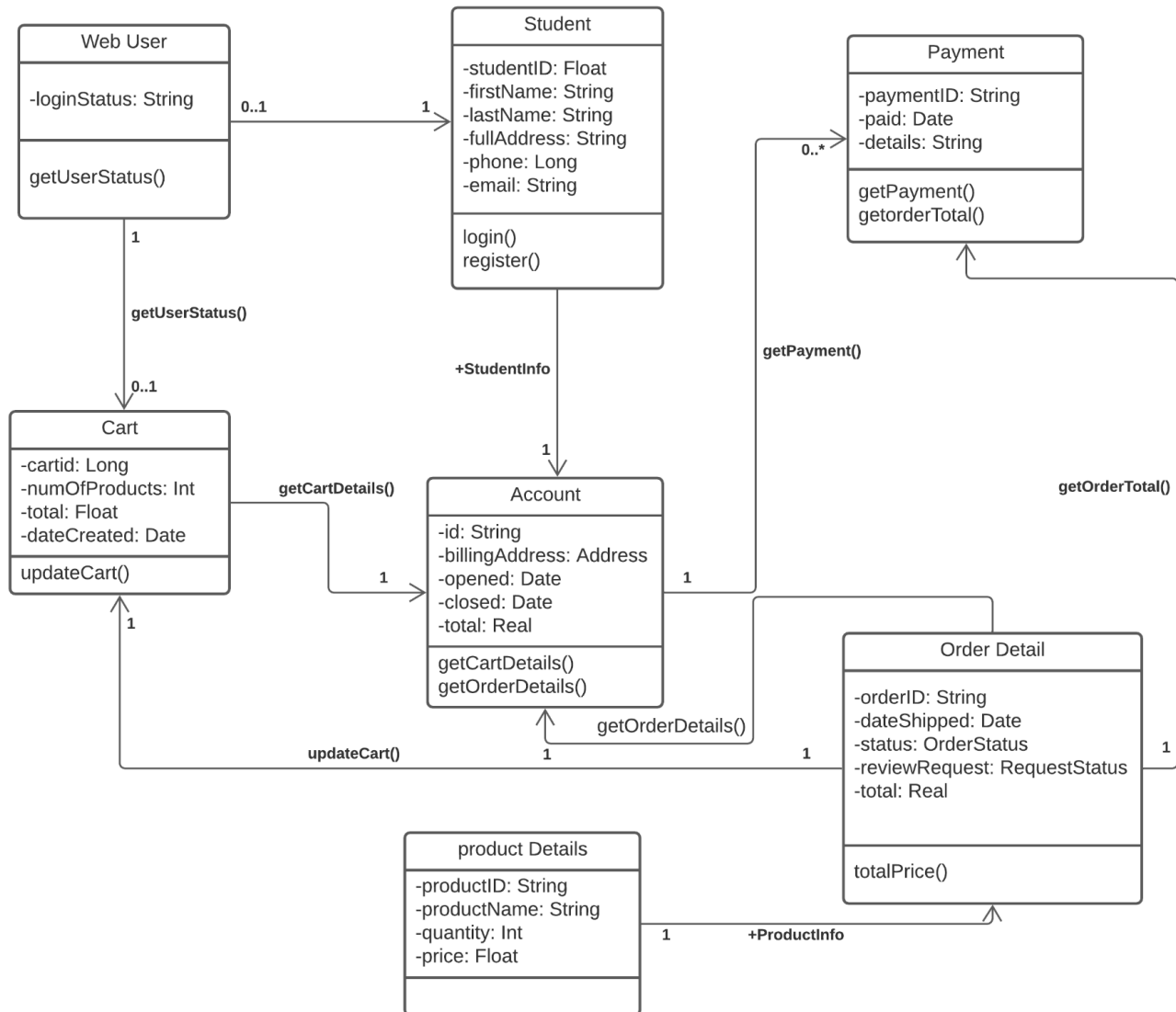
Junaid

Order Payment Detail Email

Following things were being considered while developing the system sequence diagram:

- Student browse through the home page and select clothes to buy.
- Student goes to the checkout.
- Student fill out shipping information.
- System displays price information for the clothes in the cart, including shipping information.
- Student fill in credit card information.
- System authorizes purchase.
- System confirms sales immediately.
- System sends confirmation emails to student.

(6) Design Class diagram (Entire Project):



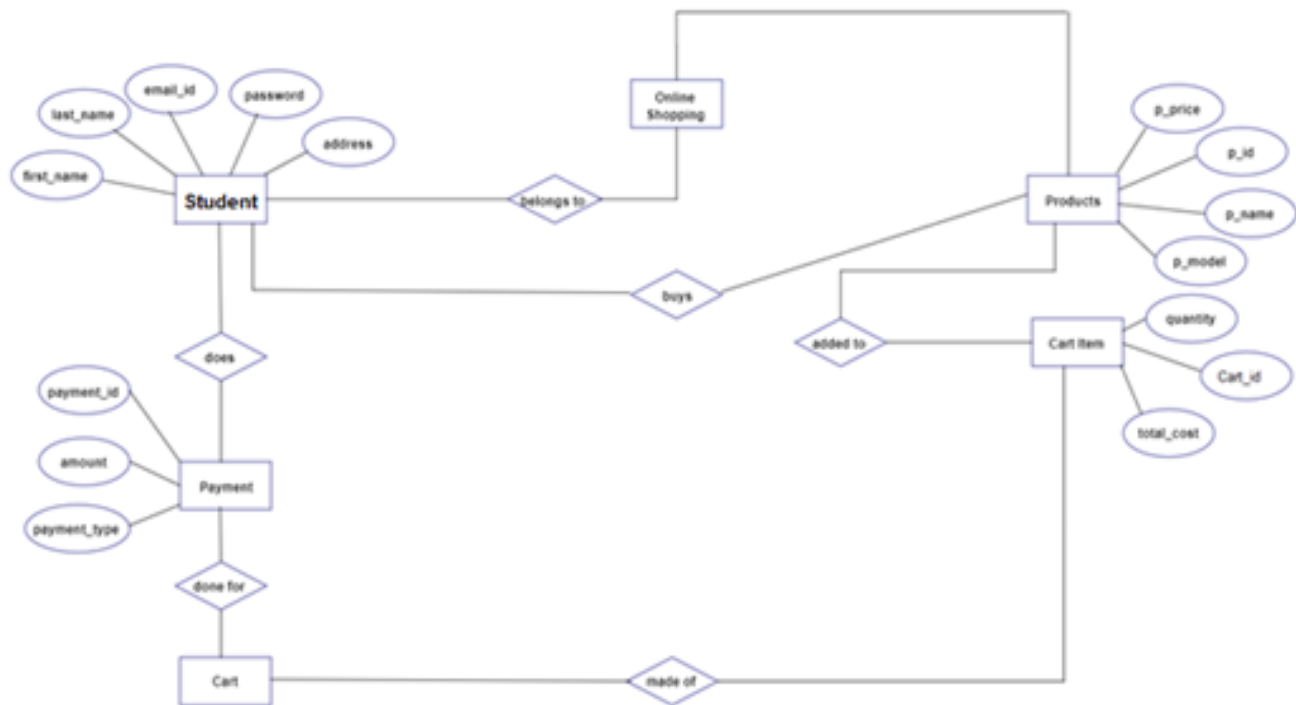
(7) Validation, Discussion and Explanation(DCD):

Above image is the design class diagram of the entire project. Some of the relations have a compositional relationship such as between the Student, Cart, Account and Order Detail class. As we know relationship is a general term covering the specific types of logical connections found on class and object diagrams. Main purpose of the diagram is to introduce some common terms such as Student, Web User, Account, Cart, Product Details, Order Details and Payment.

- **Student:** student is used to hold a reference to a Student object that allows each Payment object to be linked to the right Student object. It implements the unidirectional navigable path to the Student class which can be seen in the design class diagram.
- **Web User:** Student can browse through the website without even having to make an account and this will show the login status if the student is logged in or not and design each user's unique ID upon visiting the website.
- **Product Details:** The Order system facilitates the student to view the product details, add clothes to cart, and many more. This class provides the service to view the product details and to purchase after looking at the details of the product.
- **Order Detail:** This class has the attributes orderID which will be unique for every student once an order is placed with the dateShipped, status, reviewRequest and total, it will interact with the system directly after getting order from the student.
- **Account:** This class holds all the important information about the student account and plays an important role here because it has a compositional relationship with other 3 classes. It has attributes id which is a unique number for every account, billingAddress that will be filled by the user, open and closes which shows the date of the order created and finished respectively and total which holds the total amount of the bill to pay.
- **Payment:** payment is a class that allows the application to assign a number that uniquely identifies each Payment through paymentID. It is updated each time a new Payment object is created.
- **Cart:** Cart Class allows the website visitors to select, add, and purchase clothes from the interface, it has an attribute cartid which helps the system to analyze each cart uniquely and dateCreated attribute which shows the date it was last updated and numOfProducts and total.

CHAPTER_3 PHYSICAL DESIGN

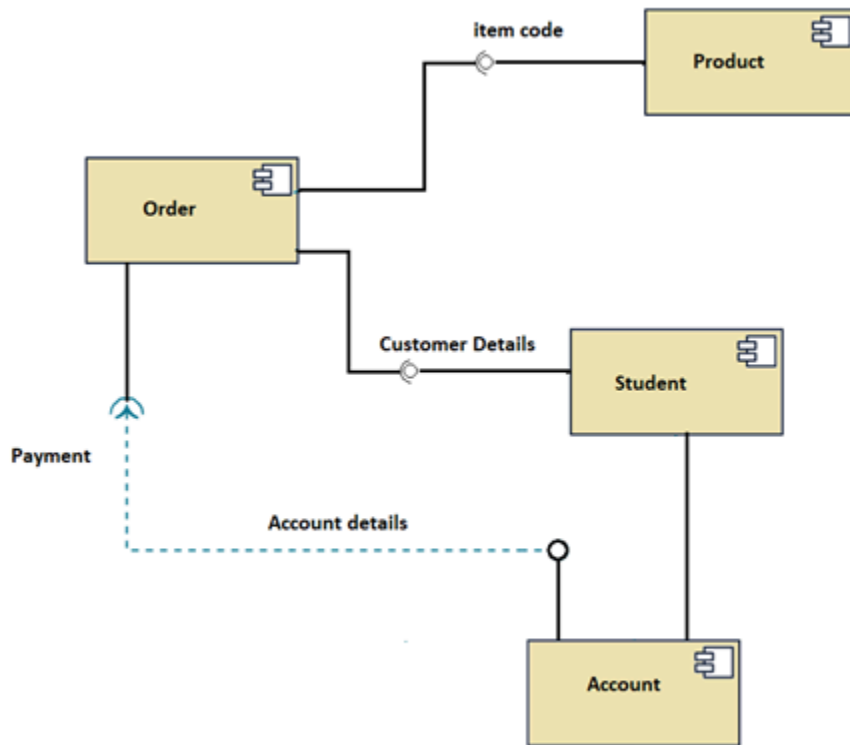
1. ERD:



Some Basic Entities:

- **Student:** It has the attributes first name, last name, email etc. This will help the system to analyse each student in a unique way
- **Payment:** It has a unique payment id, and the total amount of the items to be purchased by the student and the payment type selected by the student.
- **Cart:** Every time the student selects and adds items in the cart this will update with the product quantity, total cost, price, etc. Cart id will remain the same.
- **Products:** Every product available on the system server has a unique id with a price, model and name.

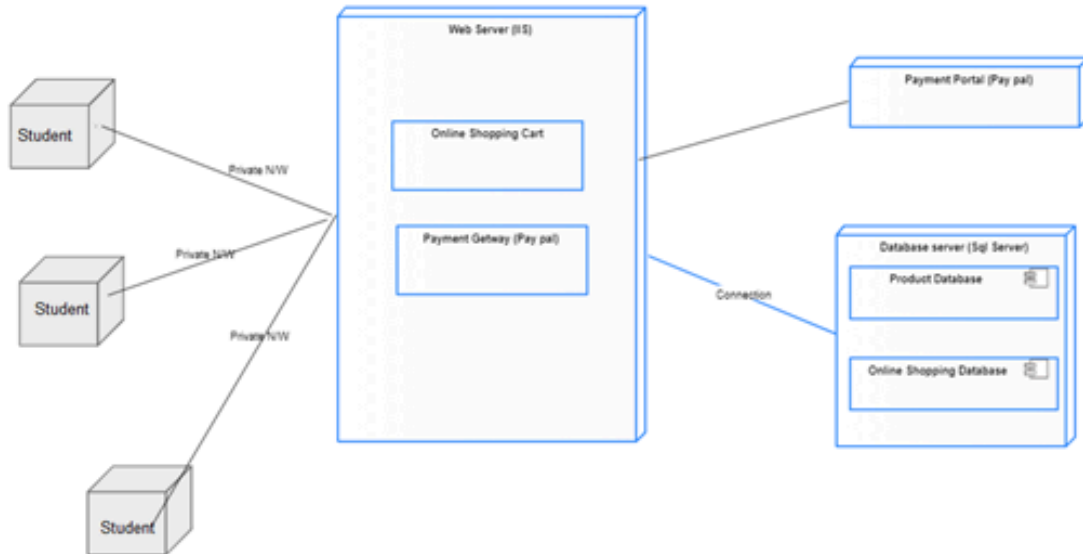
2. Component diagrams:



Four Basic Components:

- **Product:** Products that will be available in the web service for purchase
- **Order:** Whenever the user place the order after adding the product in the list
- **Student:** User who will use the web service for online shopping
- **Account:** A direct interactive entity which will get updated every time user add items to their cart

3. Deployment diagrams:



Students interact with the web server through the private network and can access the web services and web applications and further can proceed for payment gateway to pay bills, there is a separate database on SQL server which will get updated as per the activities of the user.

CHAPTER_4 VALUATION OF ANALYSIS & DESIGN

Milestones	Analysis Section					
	Points (1-5)	Experiences	Problems encountered	Unresolved problems	Project evaluation	Evaluation of UML and other tools
1	5	We brainstormed together as a group what project will best fit us and we could work with throughout the term	These were the initial stages of the project so we barely encountered any problems and also we had a template to follow so there was barely any problems	We didn't have any unresolved problems	Analysis and design of a clothing online store called a clothing web based system	No use of any tools for UML
2	4	We all had different ideas for the requirements document stories so we had to choose what story to go with	Deciding what use cases we were going to have and how many were going to be included or excluded	We resolved by going through the story together, reading the slides you provided and then deciding collectively what use cases to go with and if they were included or excluded	Developed a requirements story based on our use case models	No use of any tools for UML
3	4	We all made our individual use case descriptions and although it was tough, we were able to do it.	We had a delay in submission of milestone 3 due to personal complications.	Nothing related to the project but we had some personal complications we resolved but made us submit the milestone a few days late	We wanted to put more extended and included use case descriptions and we did in milestone five.	For the class diagram we depended on the lucid online tool. It was really great using it as it contained all the major signs needed for creating class diagrams. The only complication with it was that with the student account you can only use a certain amount of space which was limited but enough to create one diagram at a time.

4	4	We each choose a use case and create sequence diagrams and system sequence diagrams. We helped each other as well in creating the diagrams and making it perfect	We had problems making a Design Class Diagram but then after looking at zoom lectures again we were able to create DCD for the entire project.	We are a very understanding team and we worked together most of the time on zoom calls to resolve any problems before we submit the milestones	We wanted to work on Relational Database Schema to practice the stuff we got from the videos but due to time complexity we were not able to work on it.	We used Drawio to create our sequence diagrams and system sequence diagrams. It was very easy to use. One team member also used
5	5	We pulled together all our milestones and organized them into one document. It was nice to see all the work we had put together over the quarter.	We did not have any major problems while working on this milestone. It was about fixing any issues so there was no room for more.	We resolved our problems in all our previous milestones.	Despite any issues we faced during this ten week project, we were able to finish the final deliverable on time.	We used lucid chart and draw io to make corrections
AVERAGE	4.4	We are now aware of our mistakes and have learned many things from those mistakes and are ready to improve them.				

References(None)

Appendix(None)

9.1 Lessons learned

Junaid:

It was really great working on this project as a team. I was able to learn many things and gain knowledge about many new topics. This project helped me gain knowledge about working as a team and then tackling any problems along the way. With all the knowledge I have gained I'm now confident enough to design any new product. I learned how punctuality really matters while working in a team and how to understand others' problems and be patient knowing they might be going through something.

Tobi:

I was exposed more to working with various UML tools and learnt about use case modeling for the first time. It was nice working with a small group to design a system. I hope to do more of these kinds of projects to improve my skills.

Aisha:

I already know the struggles of working with a team but with this team I learned the art of communication and patience. This solved most of the problems that we encountered as a group. This project made me learn some new skills like designing models using UML diagrams. It was very nice to work as a team as we learned the material together for each milestone.

9.2 Division of the work among team members:

Most parts of the projects were done as a team. If we talk about all the major diagrams that were based on the entire project then it was a team effort to finish it. Everyone played their role to finish the project and work was divided equally. There are few items that are done individually and all that items contain the name of the team member that helped finish that.